Software Requirements Specification (SRS)

Corporate Mental Wellness & Counseling Platform (Problem Statement #14)

Name: Dhanush Shankar Reddy
SRN: PES1UG24AM083
Section: B
Version: 1.0
Date: 28-10-2026
Status: Final

Revision history

Version	Date	Author	Change summary	Approval
1.0	28-10-2026	Dhanush Shankar Reddy	First version of the SRS	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

Table of contents

1. Introduction
2. Overall description
3. External interfaces
4. System features (detailed)
5. Non-functional requirements (detailed)
6. Quality attributes & acceptance tests
7. System models and diagrams
8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This document states what the Corporate Mental Wellness and Counseling Platform has to do and how we will know it does it. It is for the team that builds and tests the system, and for the instructor marking it. Everything here traces back to the scenario in Problem Statement #14.

1.2 Scope

The platform is an anonymous counseling portal for a company. An employee can check their stress level with a short self-assessment, book a confidential session with a licensed therapist, and watch their own wellness indicators over time. The point of the design is that the employer never learns who used the service. The system covers registration, the self-assessment, booking, the personal trend view, and the crisis path when a score is severe. It does not cover how a session is actually delivered, therapist licensing checks, or billing.

1.3 Audience

Developers, QA engineers, the wellness program administrator, and the course evaluators.

1.4 Definitions

Pseudo-ID: the random identifier a therapist sees instead of a name. Band: the stress category a score falls into. SRS: Software Requirements Specification. NFR: non-functional requirement. RTM: Requirements Traceability Matrix. AES-256: the encryption used on the identity mapping. TLS: the protocol that encrypts traffic between the browser and the platform.

2. Overall description

2.1 Product perspective

The platform is a web service split into four backend services sitting behind one API gateway: an Identity and Pseudonym Service, a Self-Assessment Service, a Booking Service, and a Notification Service. Employees reach it through a Wellness Web Portal, which only ever talks to the gateway. It connects outward to the company directory for a one-time identity check, to a crisis support service, and to a notification channel for confirmations.

2.2 Major product functions

- Register an employee once and give them a pseudo-ID for everything after that.
- Run a ten-item stress self-assessment and return a score out of 40 with a band.
- Book a confidential session into a slot the therapist has published.
- Show the employee a trend of their own last six assessments.
- Raise the crisis path and offer a priority slot when a score is severe.
- Let the administrator read group-level reports that never expose a small group.

2.3 User roles and characteristics

- Corporate Employee: uses the service, wants it quick and anonymous.
- Licensed Therapist: publishes availability, sees pseudo-IDs and session notes, never names.
- Wellness Program Administrator: reads aggregate trends, manages the program.
- Supporting systems: Corporate Identity Provider, Crisis Support Service, Notification Service.

2.4 Operating environment

A browser-based portal over HTTPS, backend services deployed independently, each data-holding service with its own store. The identity mapping store is kept apart from the rest.

2.5 Constraints

The link between a corporate identity and a pseudo-ID must be stored separately and encrypted with AES-256. No report may be produced for a group of fewer than ten people. The slot list has to load quickly during peak booking periods.

3. External interface requirements

3.1 User interfaces

A web portal with a short assessment form, a slot picker for booking, and a personal trend chart. Text is readable, and the crisis helpline is shown plainly whenever a severe score appears.

3.2 Hardware interfaces

The platform needs no special hardware. Employees and therapists use their own computer or phone with a current web browser, and the services run on standard servers.

3.3 Software interfaces

- Corporate Identity Provider: checks employment once at registration, nothing after.
- Crisis Support Service: contacted on the escalation path for a severe score.
- Notification Service: sends booking confirmations through a message queue.

3.4 Communications

All traffic runs over TLS. The API gateway is the only entry point from the internet; it checks the session token on every request and passes only the pseudo-ID to the services behind it.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow FR-###.

4.1 Registration and identity

Description: Confirm once that the user is a current employee, then identify them only by a pseudo-ID.

Req ID	Requirement (shall…)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-001	The system shall verify the	Functional	High	Employee /	Only a current	Needs

Req ID	Requirement (shall…)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	employee once against the corporate directory, then issue a random pseudo-ID; therapists see the pseudo-ID and nothing else.			Privacy	employee can register, and a therapist's view shows no name. Test: TC-REG-01	Corporate Identity Provider
FR-006	The system shall use the pseudo-ID on every screen and record after registration.	Functional	High	Privacy	No screen after registration shows the corporate identity. Test: TC-REG-02	Applies to all services
FR-007	The system shall let the employee delete their account and remove their mapping entry, so past results can no longer be linked to them.	Functional	Medium	Employee	After deletion the pseudo-ID resolves to no one. Test: TC-REG-03	Identity Mapping Store

4.2 Self-assessment

Description: Score a short stress questionnaire and respond at once to a severe result.

Req ID	Requirement (shall…)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-002	The system shall present a ten-item questionnaire scored 0 to 40 and return the score with its band.	Functional	High	Employee	Ten items shown, total within 0 to 40, correct band returned. Test: TC-ASM-01, TC-ASM-02	Questionnaire agreed with the program
FR-005	The system shall show the crisis helpline and offer a priority slot when a score is severe.	Functional	High	Employee / Safety	A severe score shows the helpline and a priority slot. Test: TC-ASM-04	Needs Crisis Support Service
FR-008	The system shall store each result against the pseudo-ID only.	Functional	High	Privacy	No name is stored with a result. Test: TC-ASM-03	Assessment Store
FR-009	The system shall explain the band in plain words, as a wellness check and not a diagnosis.	Functional	Medium	Wellness Program Administrator	The result screen uses the agreed wording. Test: TC-ASM-05	

4.3 Booking

Description: Let employees book confidential sessions into the times therapists have published.

Req ID	Requirement (shall…)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-003	The system shall book only into slots the therapist has published, re-checking and reserving the slot in one transaction.	Functional	High	Employee / Therapist	No unpublished slot appears, and two employees cannot take the same slot. Test: TC-BK-01, TC-BK-02	Booking Store transaction

Req ID	Requirement (shall…)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-010	The system shall let the therapist publish, edit and withdraw availability slots.	Functional	High	Therapist	A published slot appears for booking; a withdrawn one does not. Test: TC-BK-06	
FR-011	The system shall send a confirmation to both sides after a booking.	Functional	Medium	Employee / Therapist	Both parties receive the confirmation. Test: TC-BK-04	Needs Notification Service
FR-012	The system shall let the employee reschedule or cancel a booking, releasing the old slot.	Functional	Medium	Employee	The old slot becomes free again. Test: TC-BK-05	Depends on FR-003

4.4 Wellness trend, notes and reporting

Description: Show employees their own history and therapists their session notes. The administrator sees totals only.

Req ID	Requirement (shall…)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-004	The system shall show the employee a trend of their own indicators over their last six assessments.	Functional	Medium	Employee	The last six results for that pseudo-ID are shown and nobody else's. Test: TC-TR-01	Needs stored results
FR-013	The system shall let the therapist read and add session notes by pseudo-ID.	Functional	Medium	Therapist	Notes are stored against the pseudo-ID. Test: TC-TR-02	
FR-014	The system shall let the therapist mark a session as attended or missed.	Functional	Low	Therapist	The session status updates. Test: TC-TR-03	
FR-015	The system shall give the administrator an aggregate wellness report by department and period.	Functional	High	Wellness Program Administrator	The report shows totals only and no individual rows. Test: TC-RPT-01	Blocked under ten people (NFR-002)

5. Non-functional requirements (detailed)

The NFRs below are measurable and tied to the test plan. IDs follow NFR-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
NFR-001	Slot times shall be converted correctly to the employee's timezone, and the slot list shall load in under 2 seconds at peak load.	Performance	High	90th percentile under 2 s; times match across zones. Test: TC-PERF-01, TC-TZ-01
NFR-002	The identity mapping shall be stored separately and encrypted with AES-256 at rest, and no report shall be produced for a group of fewer than ten.	Privacy	High	Mapping store is encrypted; a report for nine people is blocked. Test: TC-SEC-01, TC-RPT-02

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
NFR-003	The platform shall be available 99.9% of each month, outside planned maintenance.	Reliability	High	Monthly uptime report shows 99.9% or better. Test: Ops reports
NFR-004	A failure in one service shall not stop the others; scoring and the crisis helpline keep working when notifications are down.	Availability	High	Assessment works with the Notification Service stopped. Test: TC-RES-01
NFR-005	The portal shall meet WCAG 2.1 AA and work with a screen reader.	Usability / Accessibility	Medium	Accessibility review passes. Test: TC-UX-01

5.1 Security

5.1.1 Security objectives

- Nobody at the employer, and no therapist, can find out who used the service.
- A breach of any single data store does not reveal who an employee is.

5.1.2 Security requirements

Req ID	Requirement (shall…)	Type	Priority	Acceptance criteria / Test case ref
SR-001	The system shall use TLS 1.2 or higher on every network connection.	Security	High	A plaintext connection is refused. Test: TC-SEC-02
SR-002	The system shall allow the identity mapping to be read only through the IMappingStore interface of the Identity and Pseudonym Service.	Security	High	No other service can reach the mapping store. Test: TC-SEC-03
SR-003	The API gateway shall check the session token on every request.	Security	High	A request without a valid token gets 401. Test: TC-SEC-04
SR-004	The API gateway shall forward only the pseudo-ID to the services behind it.	Security	High	No name reaches the assessment or booking services. Test: TC-SEC-05
SR-005	The system shall keep names and questionnaire answers out of every log.	Security	High	A log review finds neither. Test: TC-SEC-06

6. Quality attributes & acceptance tests

- Exit criteria for acceptance: every high-priority functional requirement is implemented and verified, no critical NFR fails, and the RTM shows all test cases passed.
- Acceptance test suites: registration and identity, self-assessment, booking, trend and reporting, performance, security, and accessibility.

7. System models and diagrams

7.1 UML use-case diagram

Two use-case diagrams follow: one for the employee, and one for the therapist and the administrator.

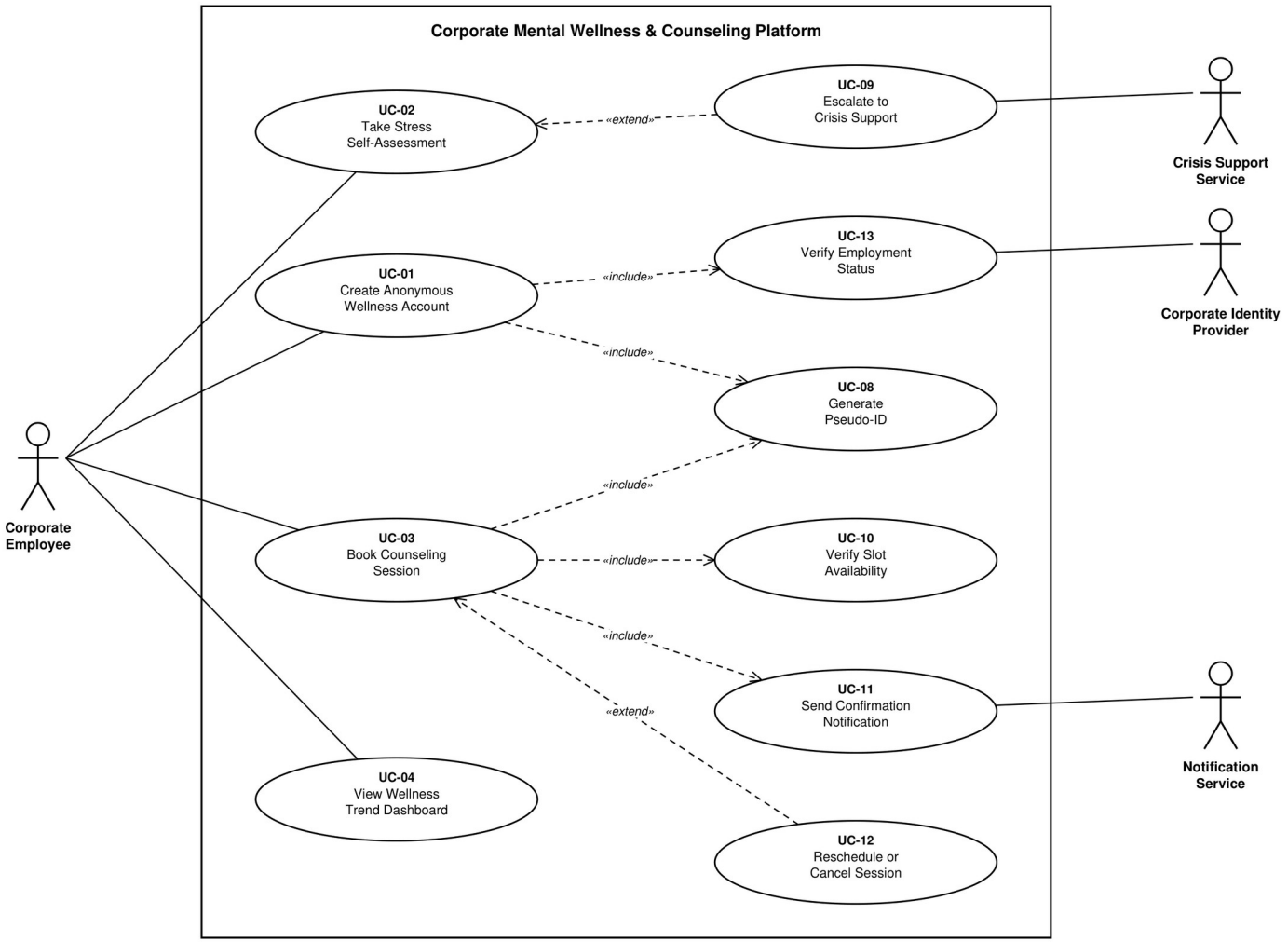


Figure 1. Use-case diagram 1: Corporate Employee view

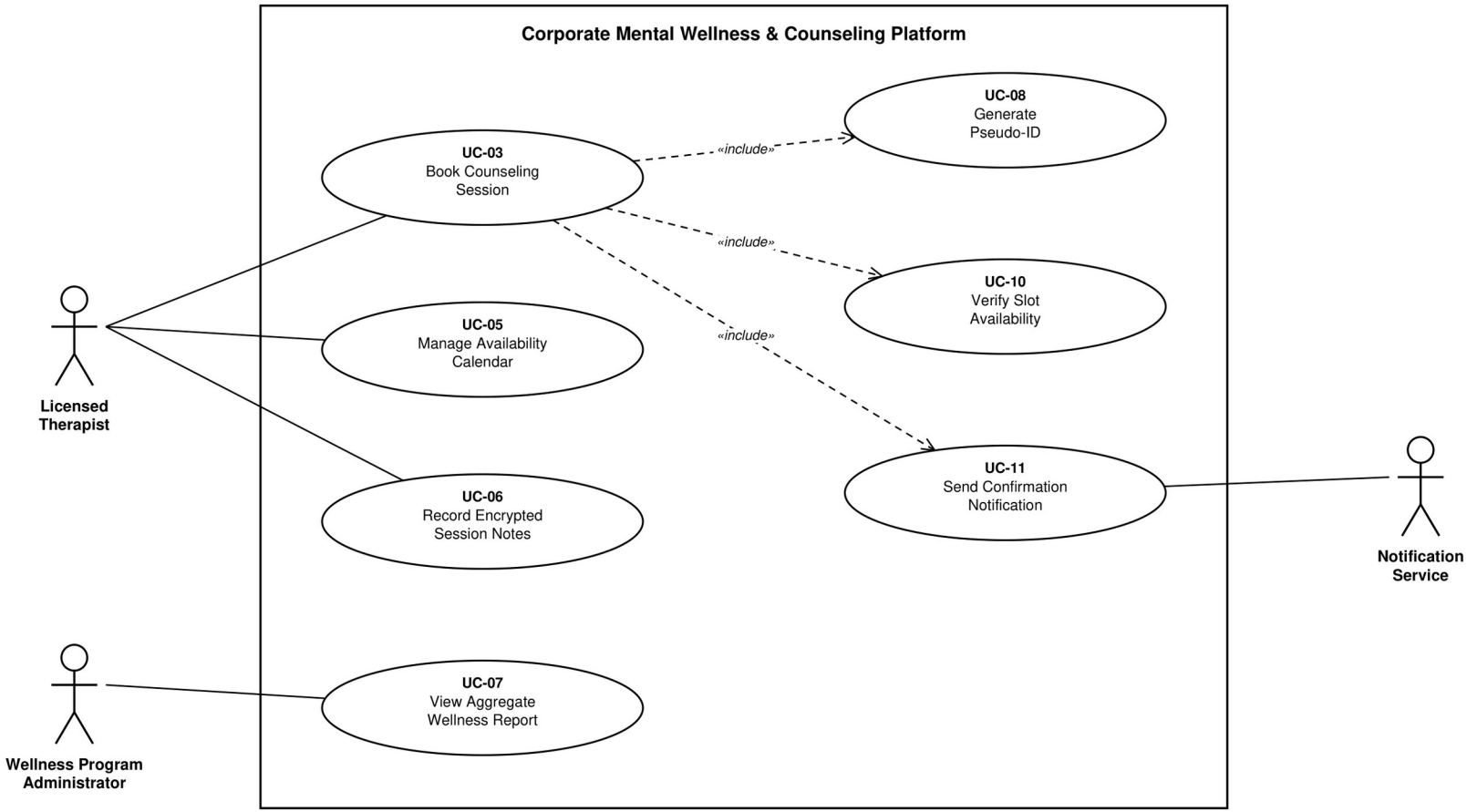


Figure 2. Use-case diagram 2: Licensed Therapist and Wellness Program Administrator view

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design spec	Module	Test case(s)	Status (N/P/A)	Comments
FR-001	Verify once, pseudo-ID	4.1 / SAD 3.4	Identity and Pseudonym	TC-REG-01	N	UC-01, UC-08, UC-13
FR-002	Ten-item assessment	4.2 / SAD 4.3	Self-Assessment	TC-ASM-01, TC-ASM-02	N	UC-02
FR-003	Book published slot	4.3 / SAD 4.3	Booking	TC-BK-01, TC-BK-02	N	UC-03, UC-10
FR-004	Six-assessment trend	4.4 / SAD 3.4	Self-Assessment	TC-TR-01	N	UC-04
FR-005	Crisis on severe score	4.2 / SAD 4.2	Self-Assessment	TC-ASM-04	N	UC-09
FR-011	Booking confirmation	4.3 / SAD 3.4	Notification	TC-BK-04	N	UC-11
FR-012	Reschedule or cancel	4.3 / SAD 3.4	Booking	TC-BK-05	N	UC-12
FR-015	Aggregate report	4.4 / SAD 3.4	Self-Assessment	TC-RPT-01	N	
NFR-001	Slot list under 2 s	5 / SAD 3.5	Booking, API Gateway	TC-PERF-01, TC-TZ-01	N	
NFR-002	Separate encrypted mapping	5 / SAD 3.9	Identity and Pseudonym	TC-SEC-01, TC-RPT-02	N	
SR-003	Token check at gateway	5.1.2 / SAD 3.9	API Gateway	TC-SEC-04	N	